UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

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|---|---|---|
| COURSE | : | INTERACTIVE MULTIMEDIA |
| COURSE CODE | : | CSC413 |
| EXAMINATION | : | NOV 2026 (MOCK) |
| TIME | : | 3 HOURS |

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

*This marking scheme consists of 6 printed pages*

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## PART A - ANSWER KEY

| Q | Ans | Statement | Explanation |
|---|---|---|---|
| A1 | TRUE | A web browser displays hypertext documents. | Browsers render web pages. |
| A2 | TRUE | Authoring tools help developers organise and edit multimedia elements. | Authoring tools assemble media. |
| A3 | TRUE | A scanner is an input device. | A scanner captures documents into the computer. |
| A4 | TRUE | Hypermedia combines multimedia with hyperlinks. | Hypermedia adds linked navigation to multimedia. |
| A5 | FALSE | A vector image becomes pixelated when enlarged. | Vector images scale without pixelation. |
| A6 | TRUE | A storyboard is part of project planning. | Storyboards plan the sequence. |
| A7 | TRUE | Interactivity gives users control over content and flow. | Interactivity is a defining feature of many multimedia products. |
| A8 | FALSE | 3D animation does not use modelling. | 3D animation relies on models. |
| A9 | FALSE | MIDI files store digitized audio samples. | MIDI stores musical instructions, not samples. |
| A10 | TRUE | A storyboard is a visual plan of a multimedia sequence. | Storyboards show the planned flow. |
| A11 | TRUE | USB is a plug-and-play connection standard. | USB devices are recognised automatically. |
| A12 | FALSE | Interactivity level is irrelevant when choosing an authoring tool. | The required interactivity affects tool choice. |
| A13 | FALSE | A plotter is an input device. | A plotter produces output. |
| A14 | TRUE | A prototype is an early model of a product. | Prototypes test ideas before production. |
| A15 | FALSE | A multimedia project always uses every media element. | A project selects the elements it needs. |

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## PART B - ANSWER KEY

| Q | Ans | Term | Definition |
|---|---|---|---|
| B1 | H | Multimedia | integration of text, graphics, audio, video and animation |
| B2 | C | Keyframe | important frame in an animation |
| B3 | E | Sampling | converting sound into digital values |
| B4 | D | Tweening | generating in-between frames |
| B5 | I | Linear multimedia | presentation without user control of sequence |
| B6 | M | Cel animation | hand-drawn frame-by-frame animation |
| B7 | G | Hypermedia | linked multimedia elements |
| B8 | K | Path animation | moving an object along a route |
| B9 | N | MIDI | standard for musical notes and instructions |
| B10 | B | Virtual reality | simulated environment |
| B11 | F | Frame rate | frames displayed per second |
| B12 | J | Chroma key | replacing a colour background |
| B13 | A | Bitmap | image made of pixels |
| B14 | O | Browser | software that displays web pages |
| B15 | L | Vector | image made of mathematical objects |

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## PART C - ANSWER KEY

| Q | Ans | Correct answer |
|---|---|---|
| C1 | B | Smooth jagged edges |
| C2 | C | Compiler |
| C3 | C | Draws alternate lines of a frame |
| C4 | A | Number of pixels in an image |
| C5 | A | A video compression family |
| C6 | C | Messages with a beginning and end |
| C7 | B | WAV |
| C8 | A | Permission to use a work |
| C9 | A | Frequency |
| C10 | B | Card-based |
| C11 | A | Drawing input |
| C12 | B | A visual sequence plan |
| C13 | B | Play while downloading |
| C14 | D | Loudness |
| C15 | D | Text, graphics, audio, video and animation |
| C16 | D | Another resource |
| C17 | C | Screen display |
| C18 | D | Automatic device recognition |
| C19 | C | Printed output |
| C20 | D | Drawing each frame by hand |
| C21 | D | Scanner |
| C22 | B | Frames per second |
| C23 | B | Pixels |
| C24 | C | Display web pages |
| C25 | D | Along a predefined route |
| C26 | D | Space between individual character pairs |
| C27 | D | Merging of media and computing |
| C28 | B | It combines visuals and sound |
| C29 | C | Small strokes at character ends |
| C30 | A | Links to other content |

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# PART D - WORKED ANSWERS

## QUESTION 1

*Topic: graphics*

A product catalogue must show sharp images on screen and in print.

**a) State TWO differences between bitmap and vector images.**

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

*(2 marks)*

**b) Describe TWO factors that affect image quality.**

Resolution and colour depth both affect how much detail an image can show.

*(4 marks)*

**c) Explain how colour model and resolution affect screen and print output.**

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

*(4 marks)*

## QUESTION 2

*Topic: web hypermedia*

A museum wants a website that presents its collection and lets visitors explore related exhibits.

**a) State TWO differences between linear and non-linear navigation.**

Linear follows a fixed sequence; non-linear lets users choose their own path.

*(2 marks)*

**b) Describe TWO roles of hyperlinks in the website.**

They connect related exhibits and let users move between pages.

*(3 marks)*

**c) Explain how streaming and compression affect multimedia delivery on the web.**

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

*(5 marks)*

## QUESTION 3

*Topic: animation*

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

**a) State TWO animation file formats suitable for this project.**

GIF and MP4 are suitable; SWF is also acceptable.

*(3 marks)*

**b) Describe the steps of cel animation and ONE advantage of using it.**

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

*(3 marks)*

**c) Differentiate 2-D and 2.5-D animation with ONE example for each.**

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

*(4 marks)*

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## QUESTION 4

*Topic: authoring interactivity*

A company wants an interactive product demonstration that runs on a kiosk and on the web.

**a) State TWO features to consider when choosing authoring software.**

Interactivity support and export formats are important features.

*(3 marks)*

**b) Explain the interactivity level suitable for the demonstration.**

A medium level with menus and buttons lets users explore without overwhelming them.

*(3 marks)*

**c) Explain TWO authoring tool categories appropriate for this project.**

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

*(4 marks)*

*END OF MARKING SCHEME*

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